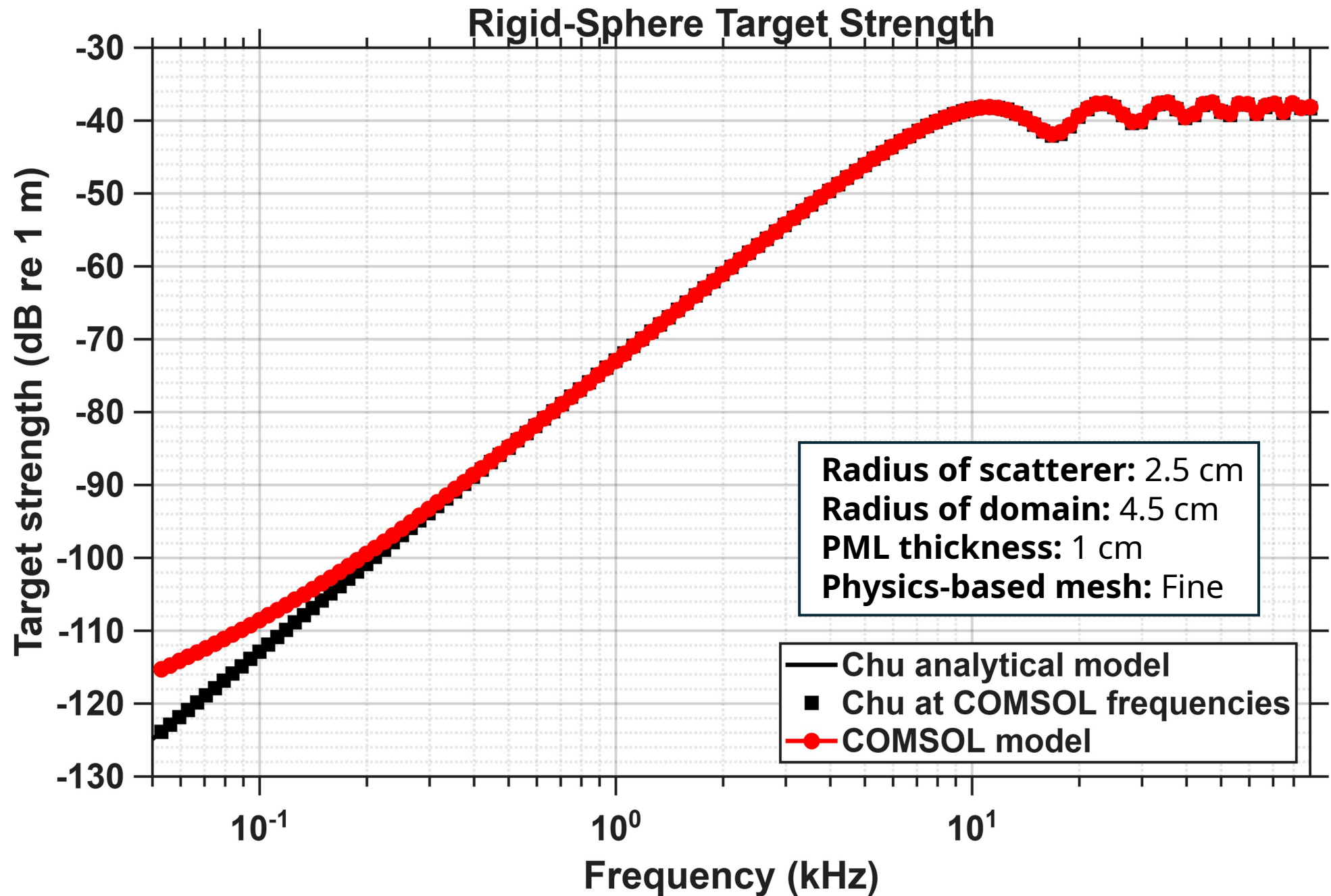
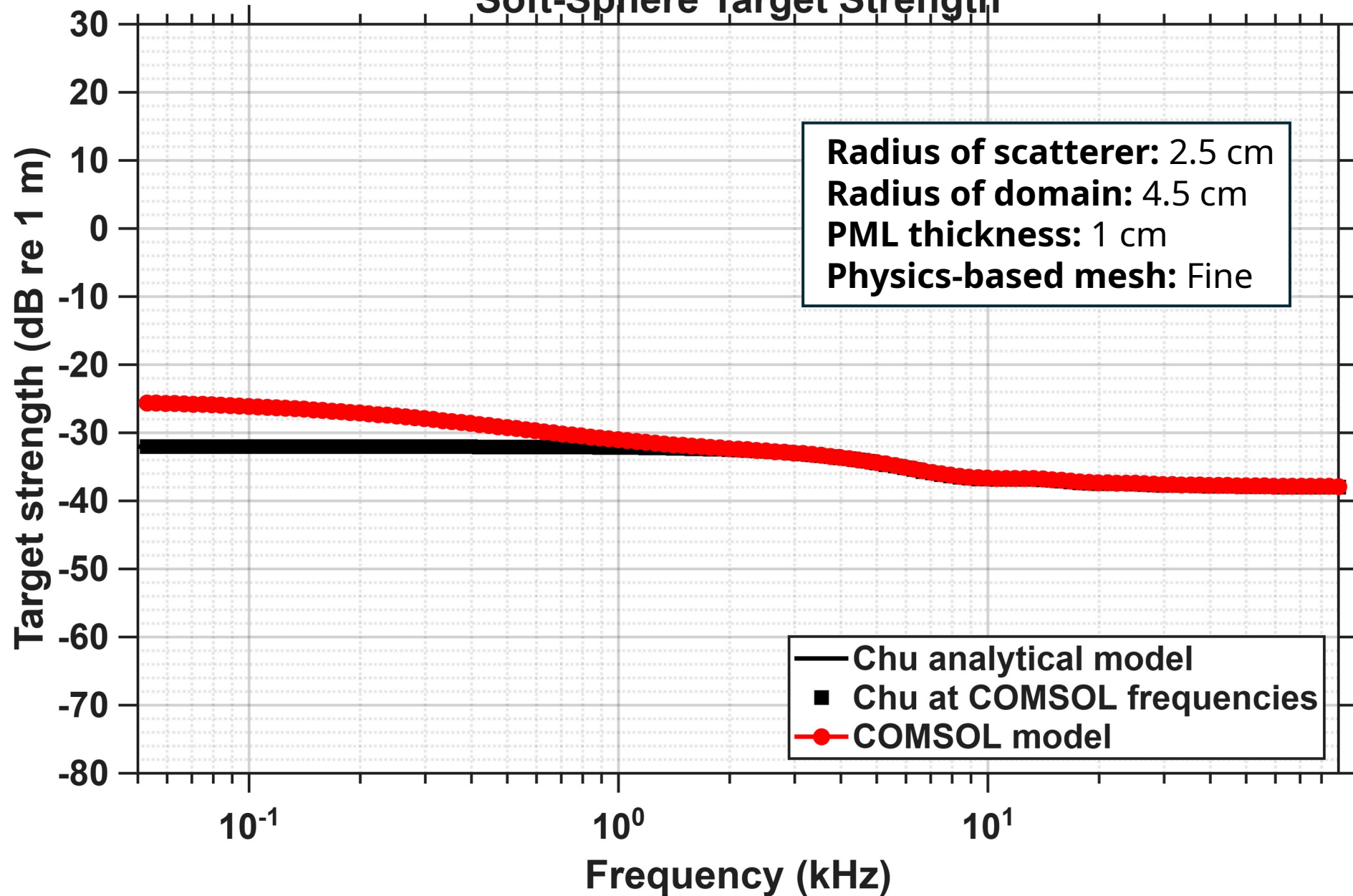


Acoustic scattering by spheres and swim bladders

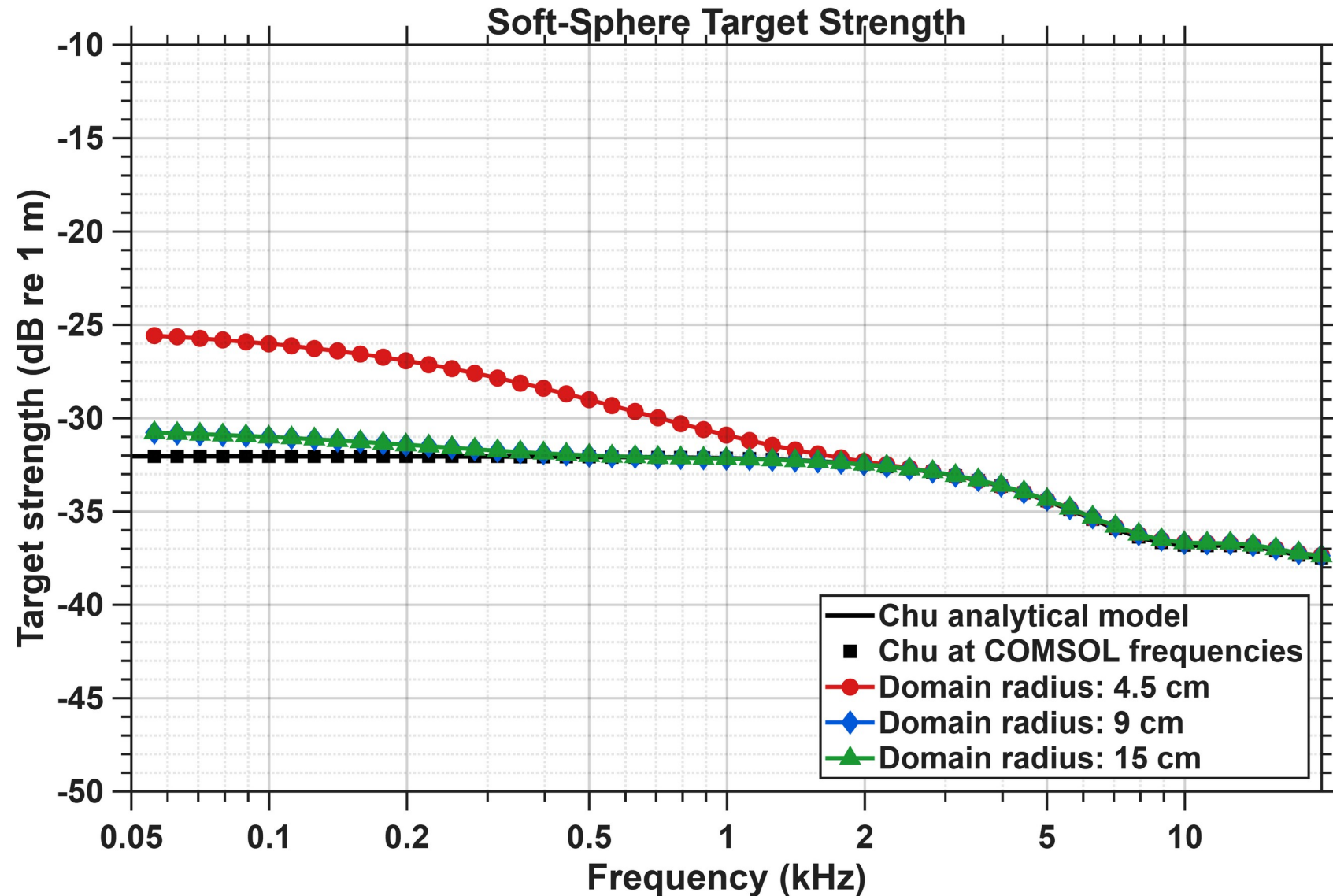
Spherical benchmark models



Soft-Sphere Target Strength



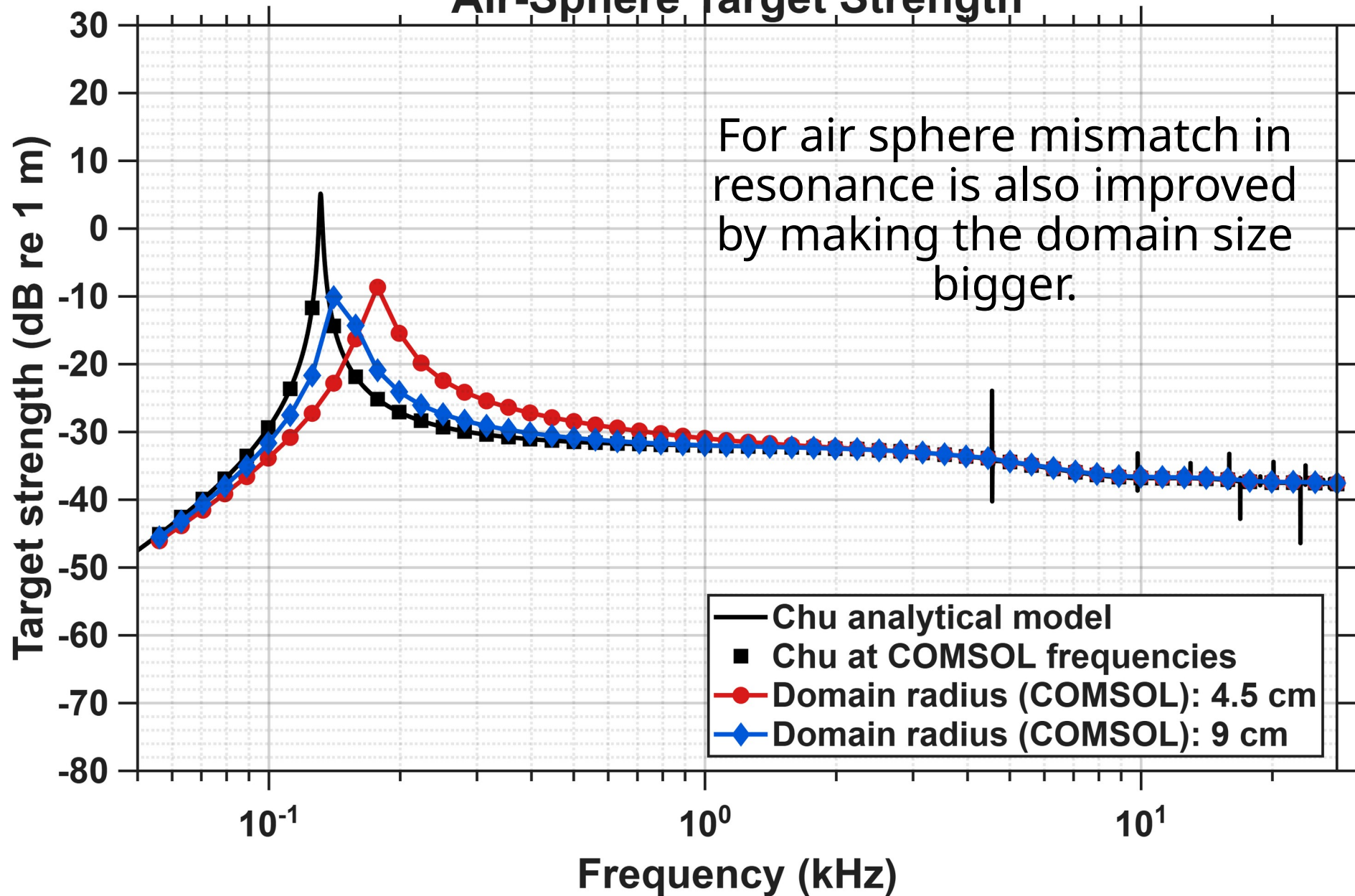
Low frequency mismatches are 'mostly' resolved by making the domain size bigger



Note:

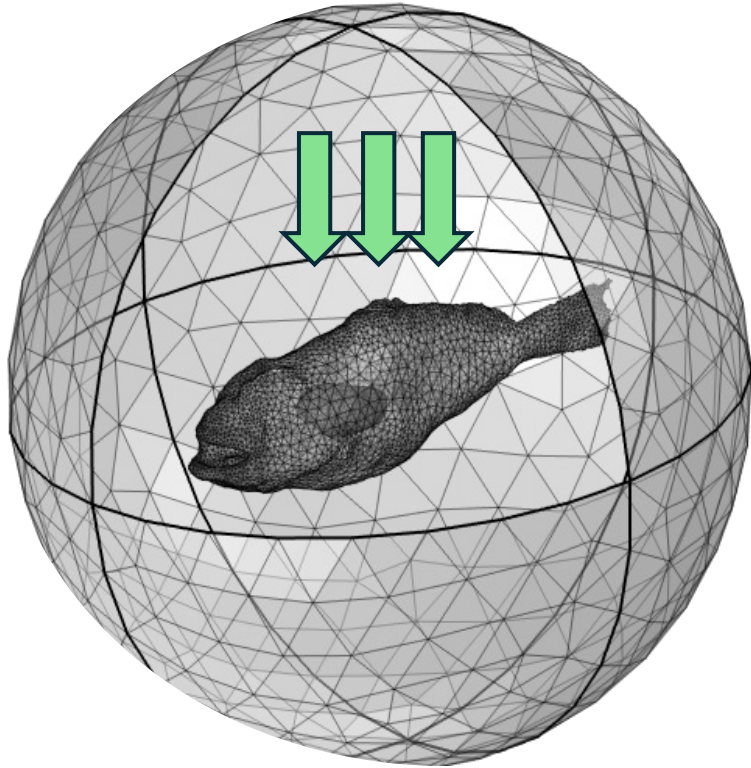
- 1) The limits on the y-axis are different compared to the previous soft sphere plot to see differences more clearly
- 2) These simulations only focus on the low frequency region where the mismatches were

Air-Sphere Target Strength

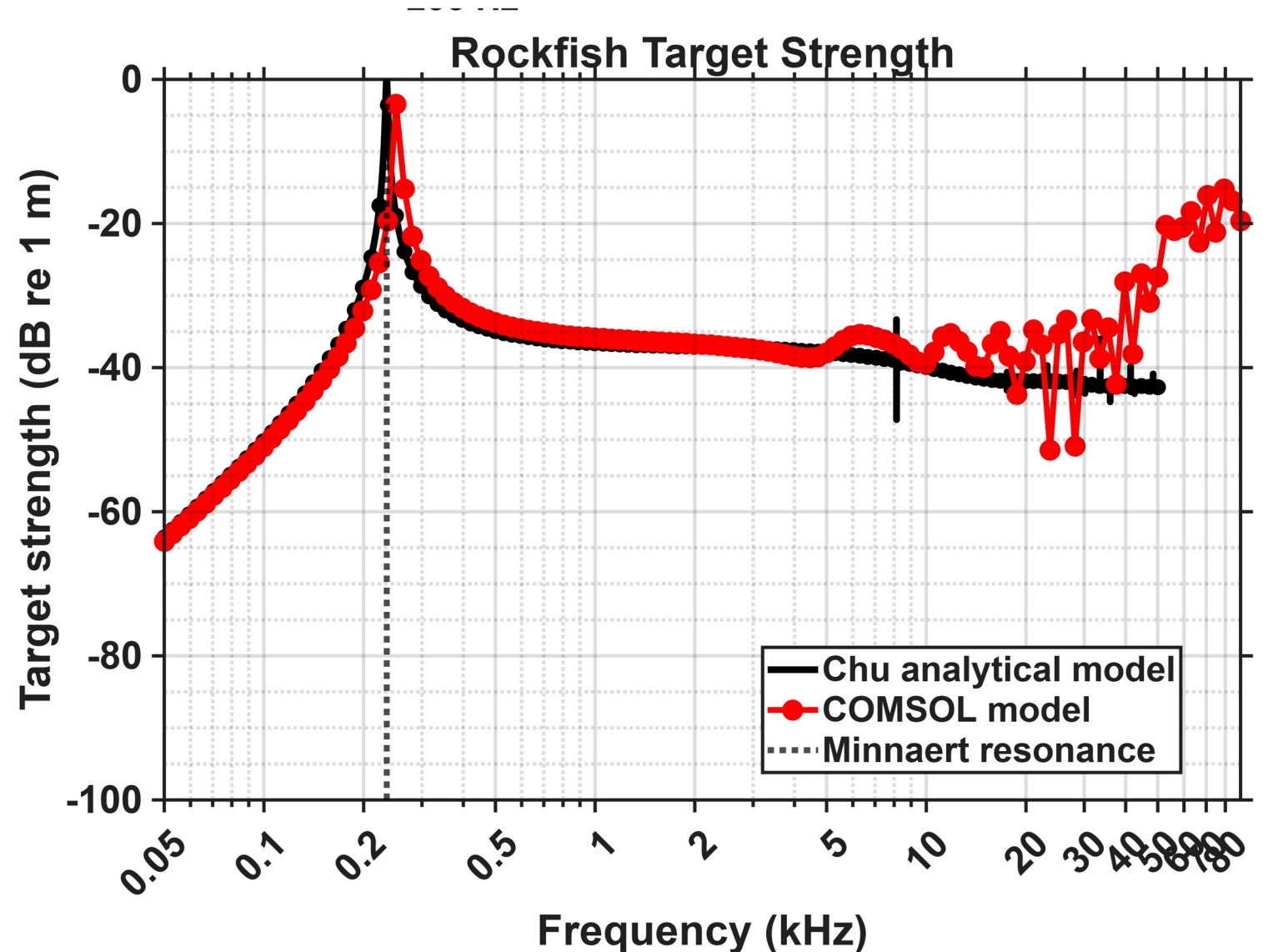


Fish and swim bladder models

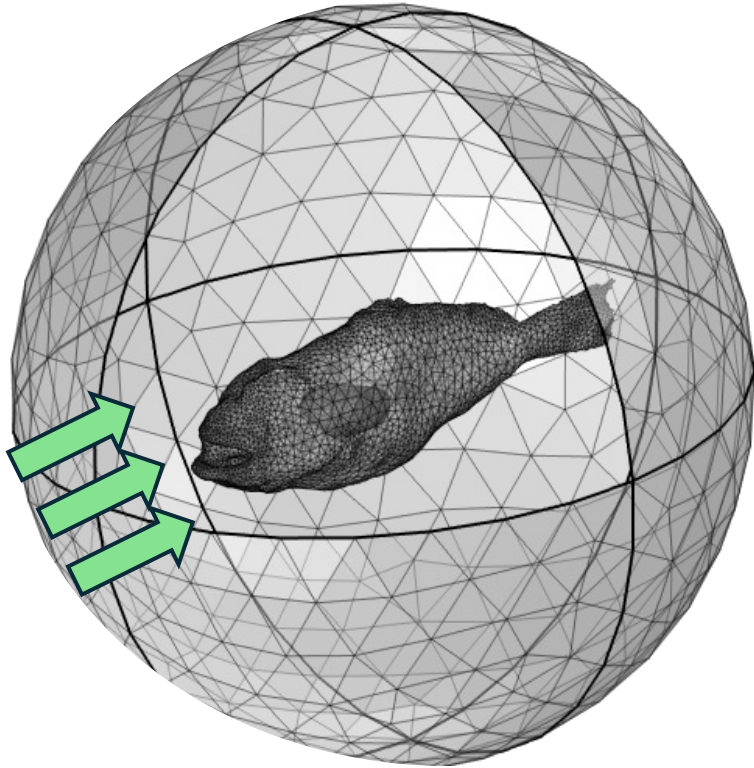
Rockfish: Dorso-ventral incidence



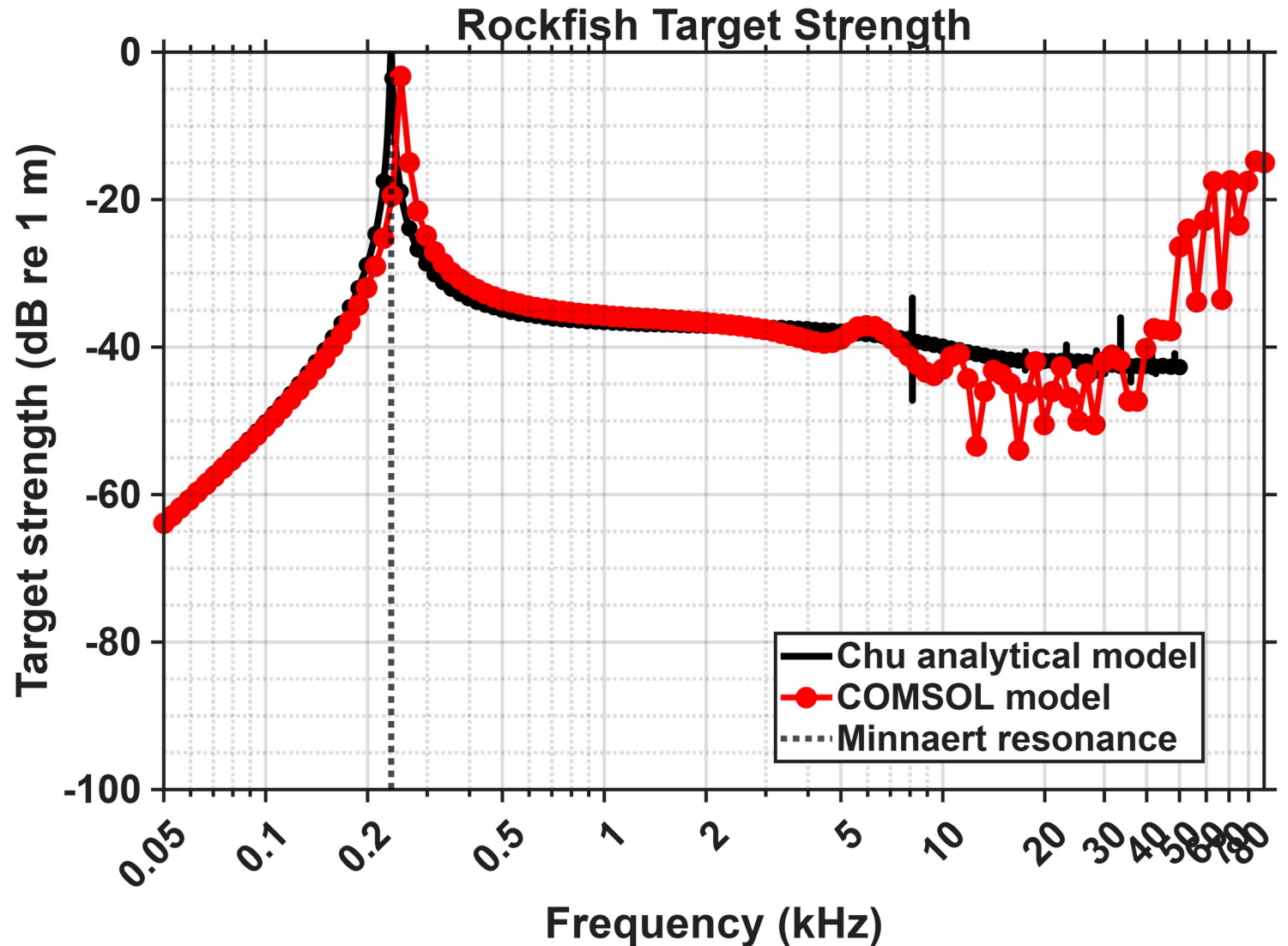
Water domain radius: 15 cm
PML thickness: 2.5 cm
Volume-equivalent spherical radius of swim bladder: 1.4 cm



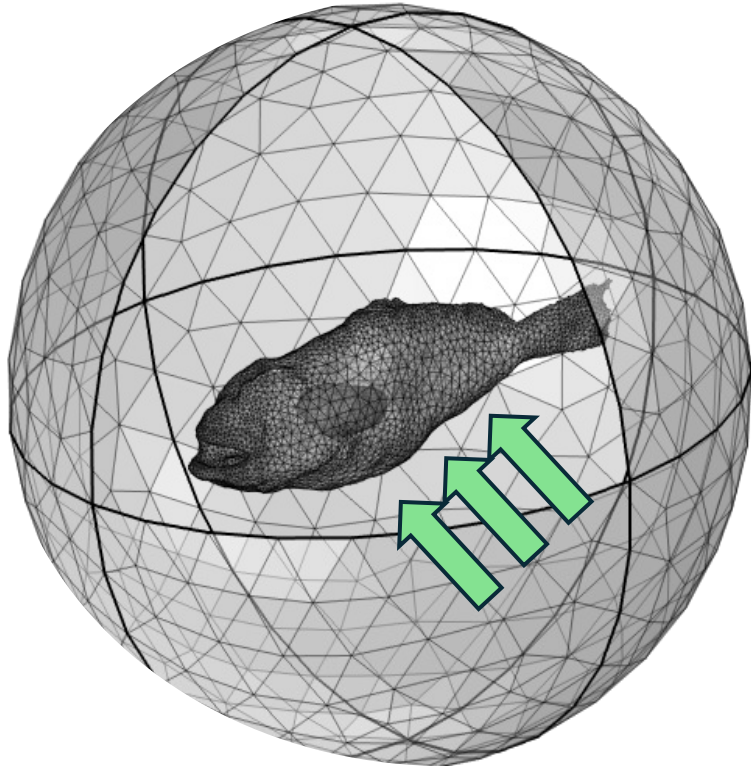
Rockfish: Anterio-posterior incidence



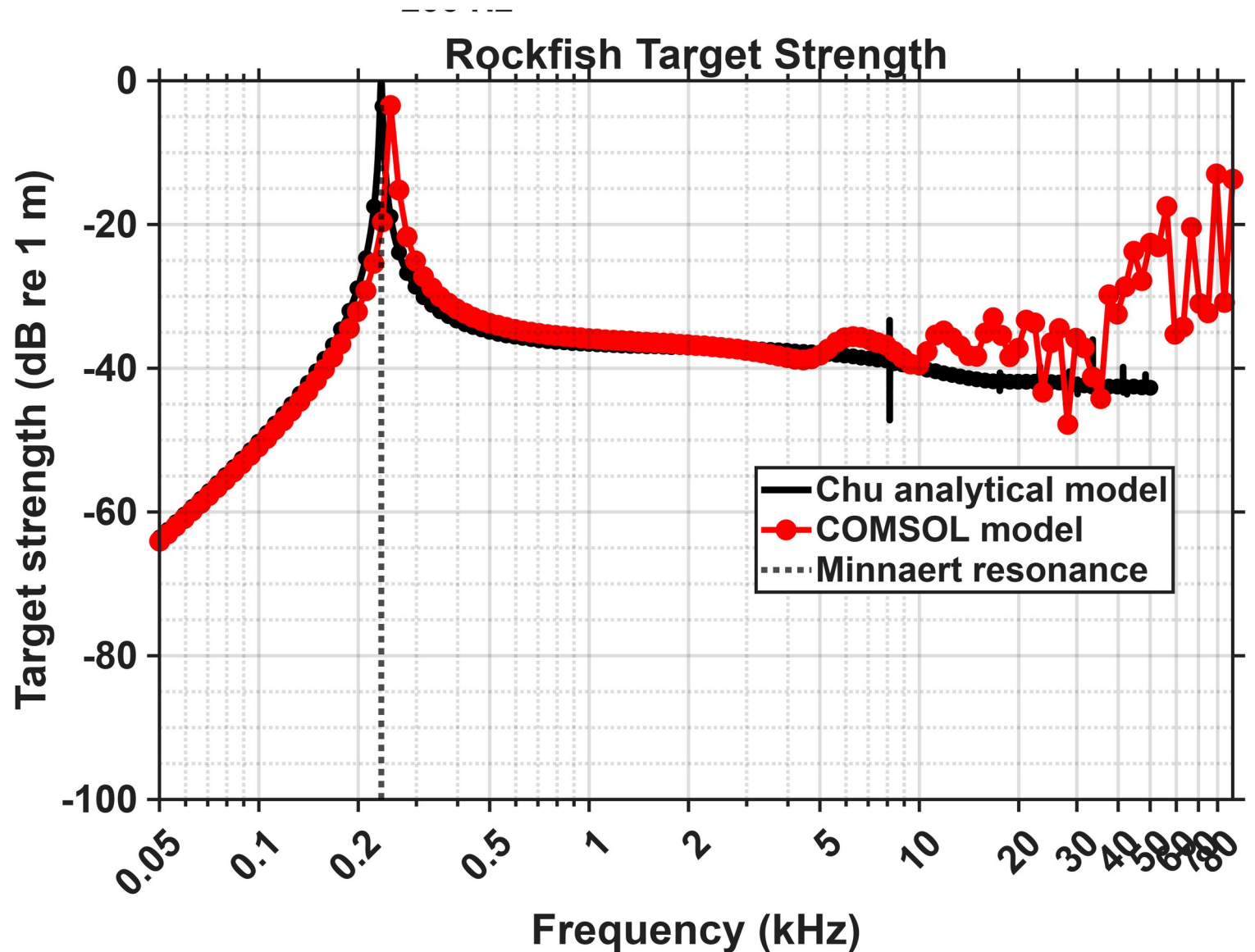
Water domain radius: 15 cm
PML thickness: 2.5 cm
Volume-equivalent spherical radius of swim bladder: 1.4 cm



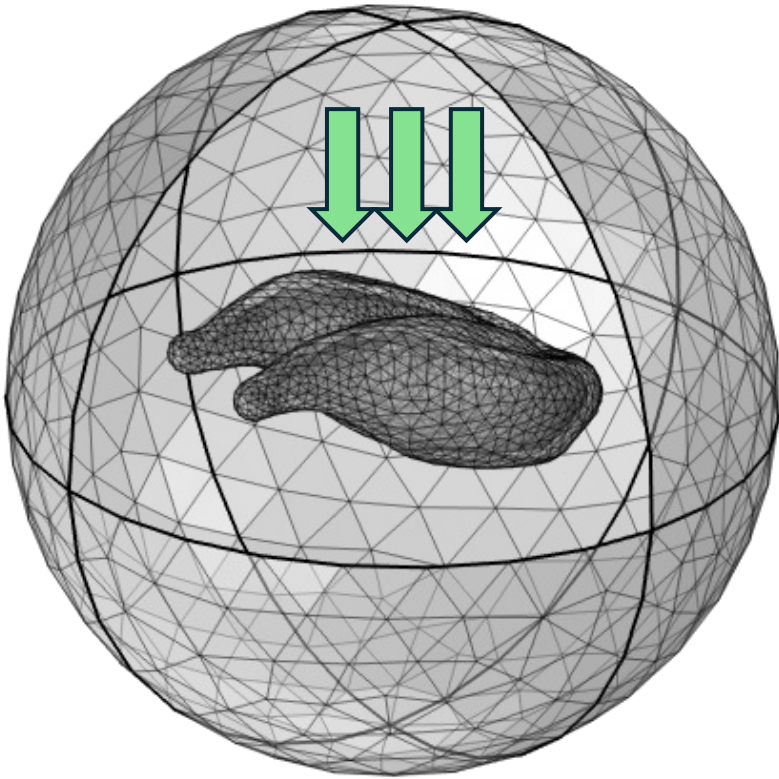
Rockfish: Lateral incidence



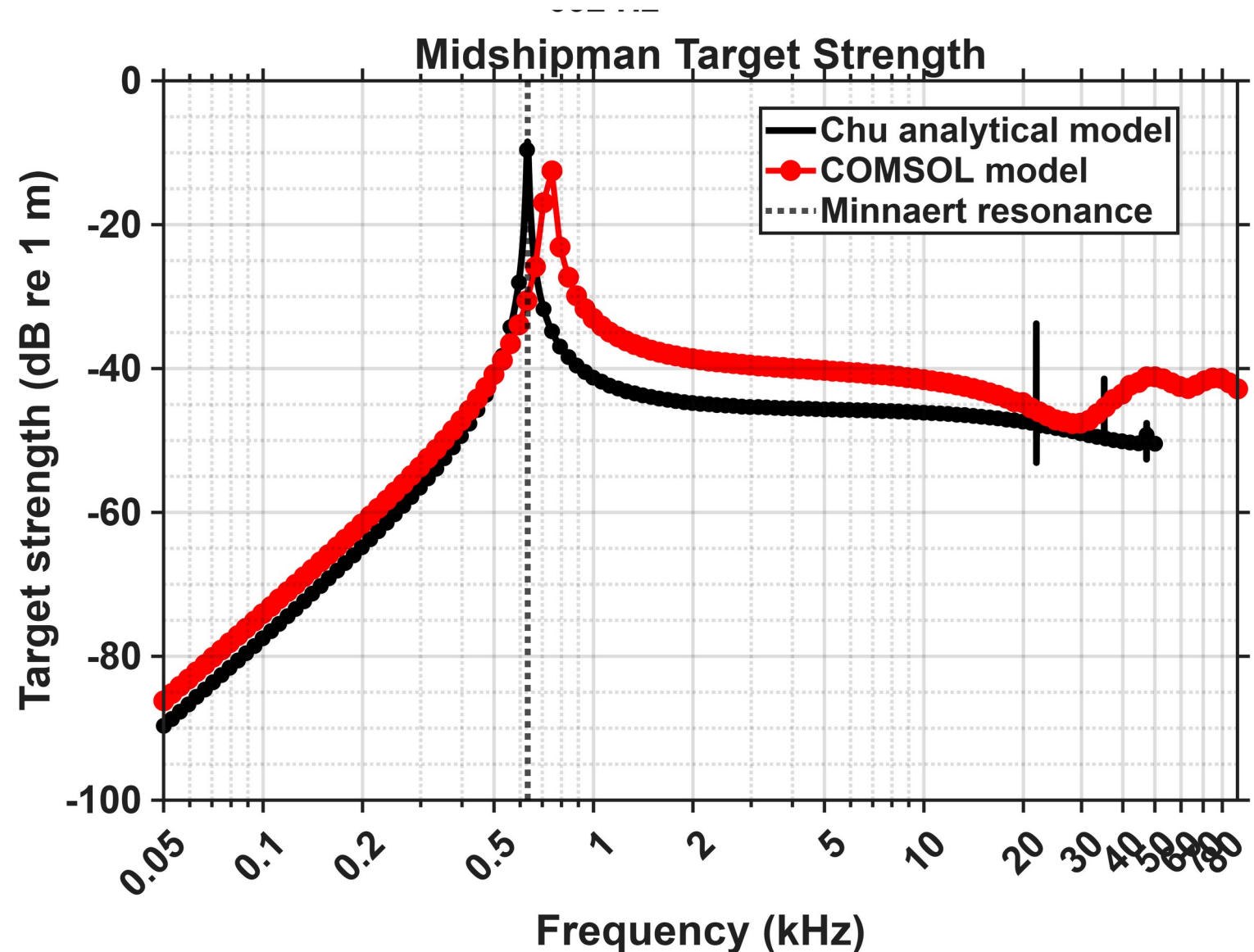
Water domain radius: 15 cm
PML thickness: 2.5 cm
Volume-equivalent spherical radius of swim bladder: 1.4 cm



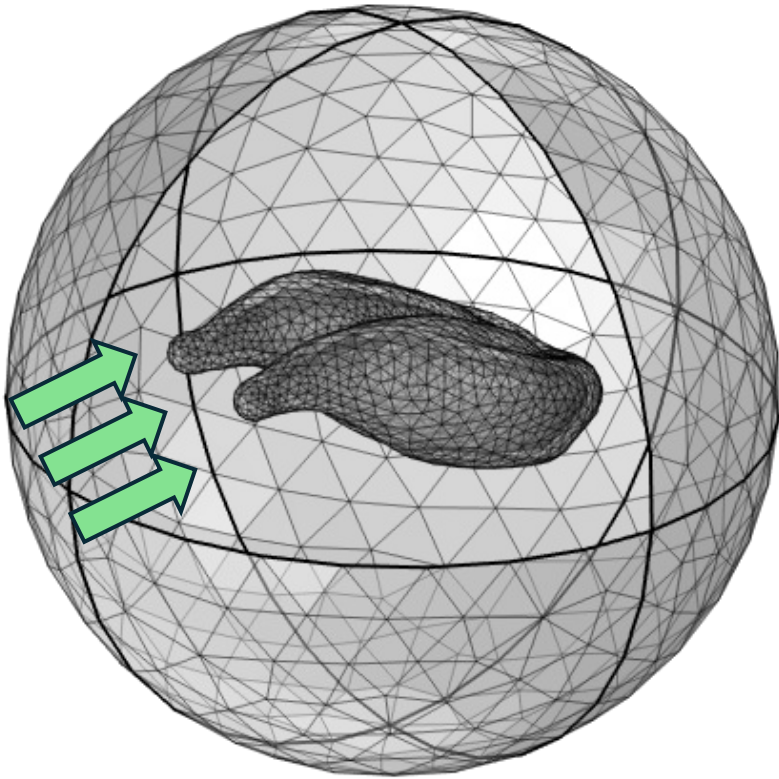
Midshipman: Dorso-ventral incidence



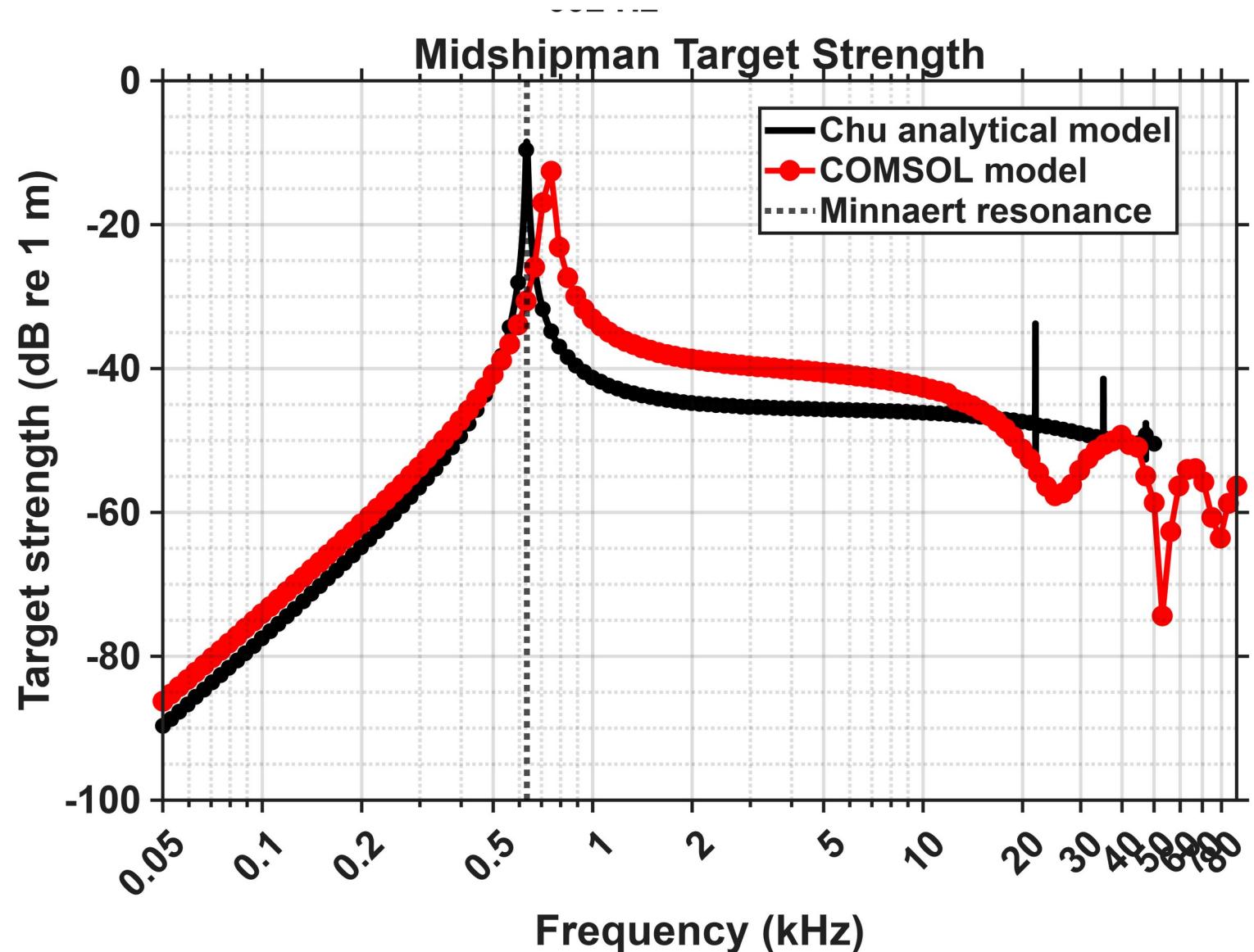
Water domain radius: 2 cm
PML thickness: 0.5 cm
Volume-equivalent spherical radius of swim bladder: 0.52 cm



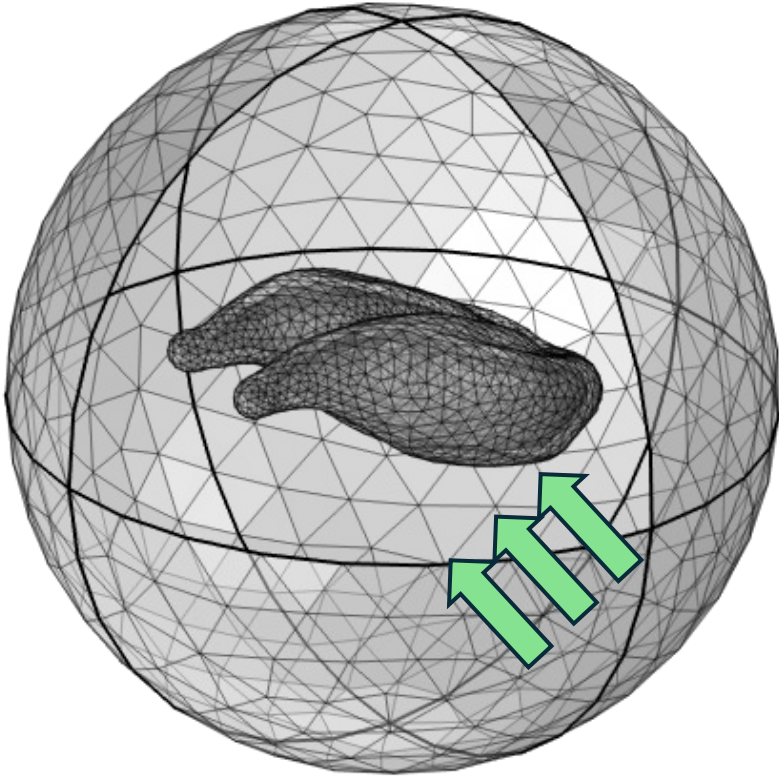
Midshipman: Anterio-posterior incidence



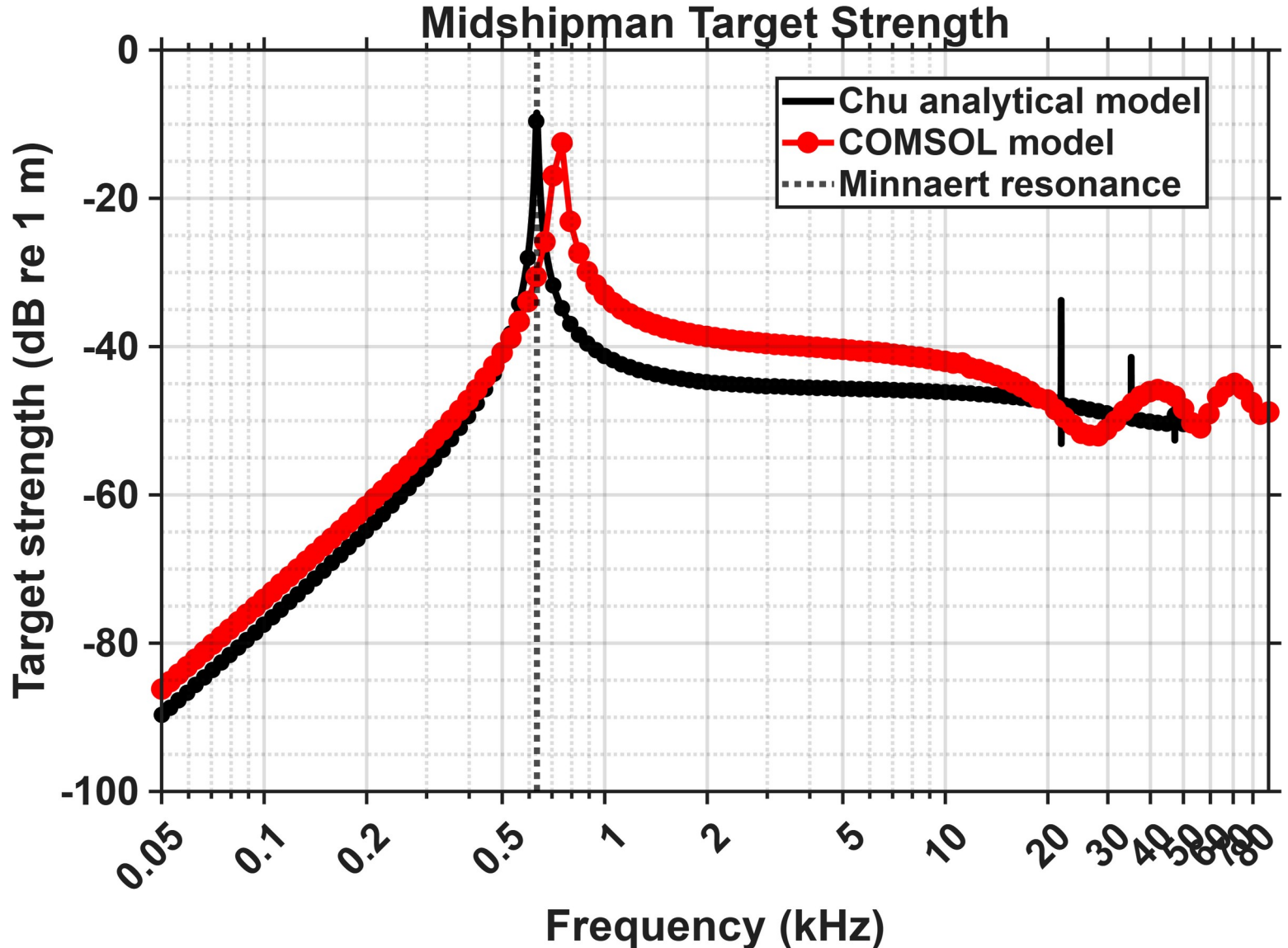
Water domain radius: 2 cm
PML thickness: 0.5 cm
Volume-equivalent spherical radius of swim bladder: 0.52 cm



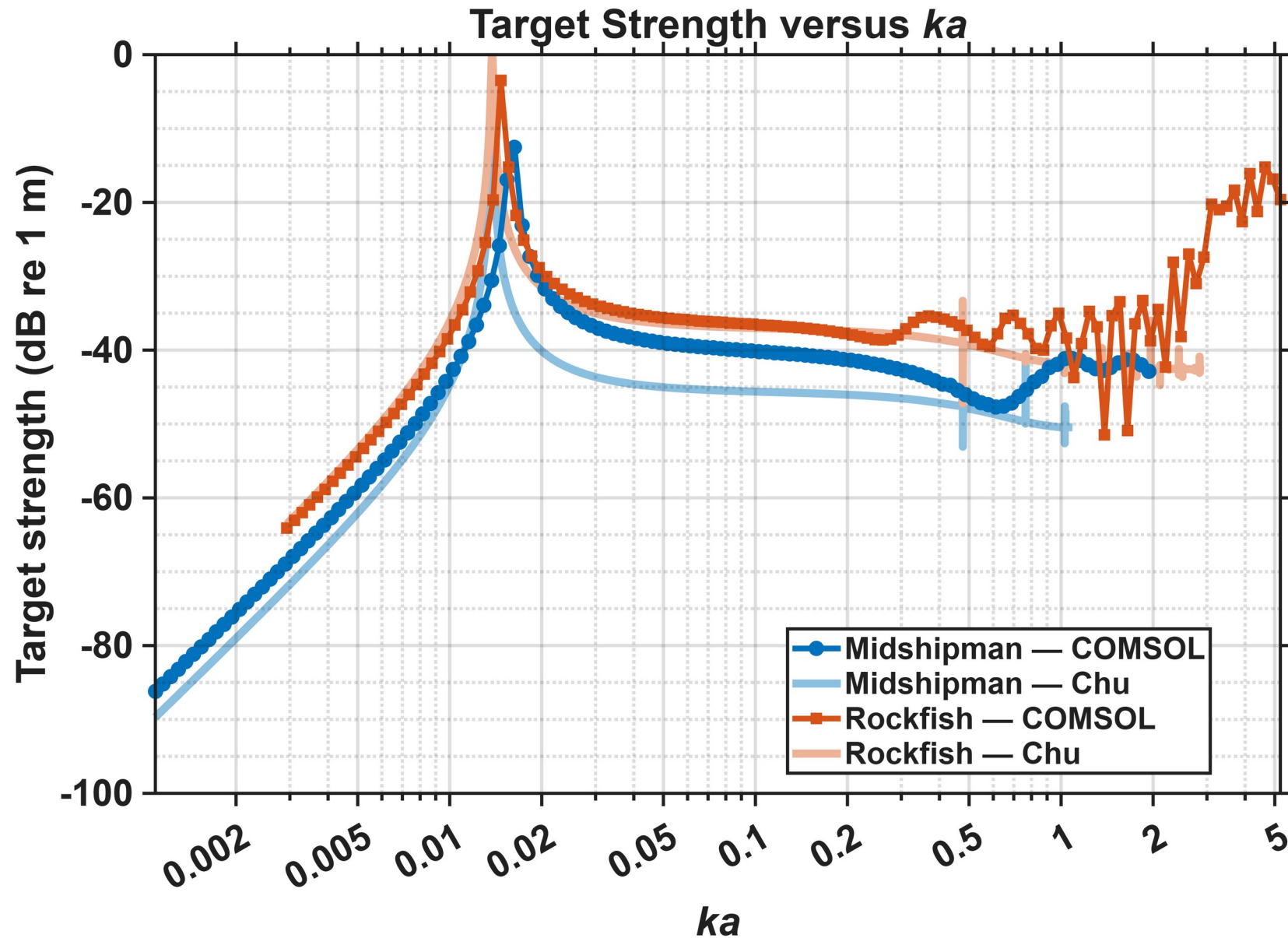
Midshipman: Lateral incidence



Water domain radius: 2 cm
PML thickness: 0.5 cm
Volume-equivalent spherical radius of swim bladder: 0.52 cm

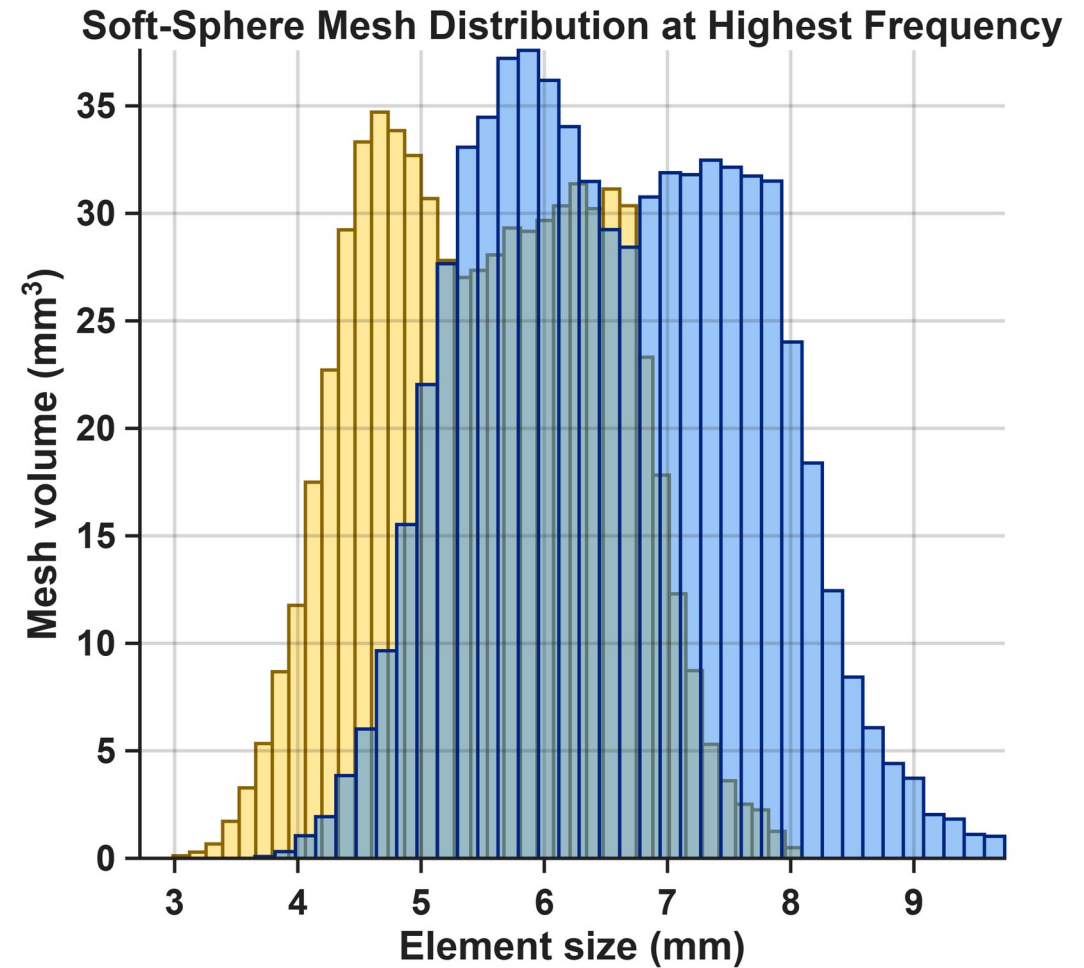
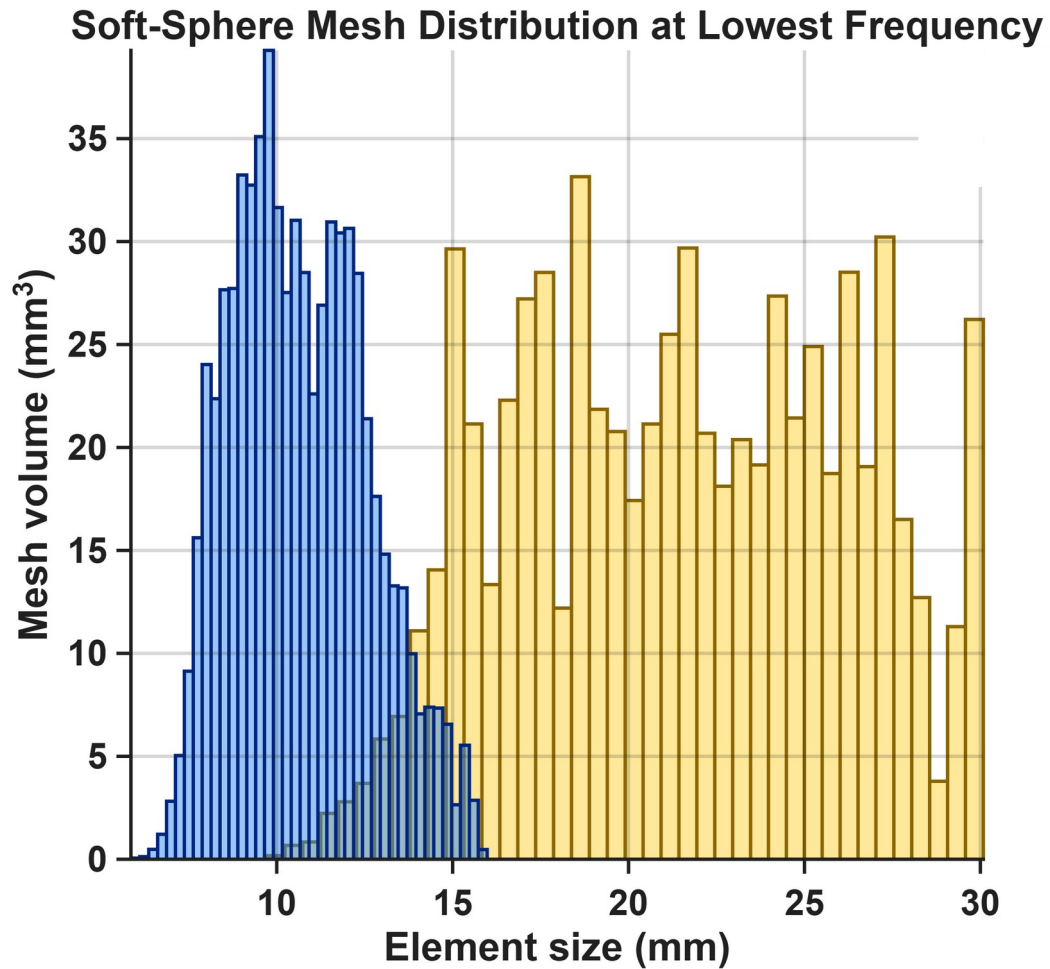


Midshipman-Rockfish dorsoventral incidence

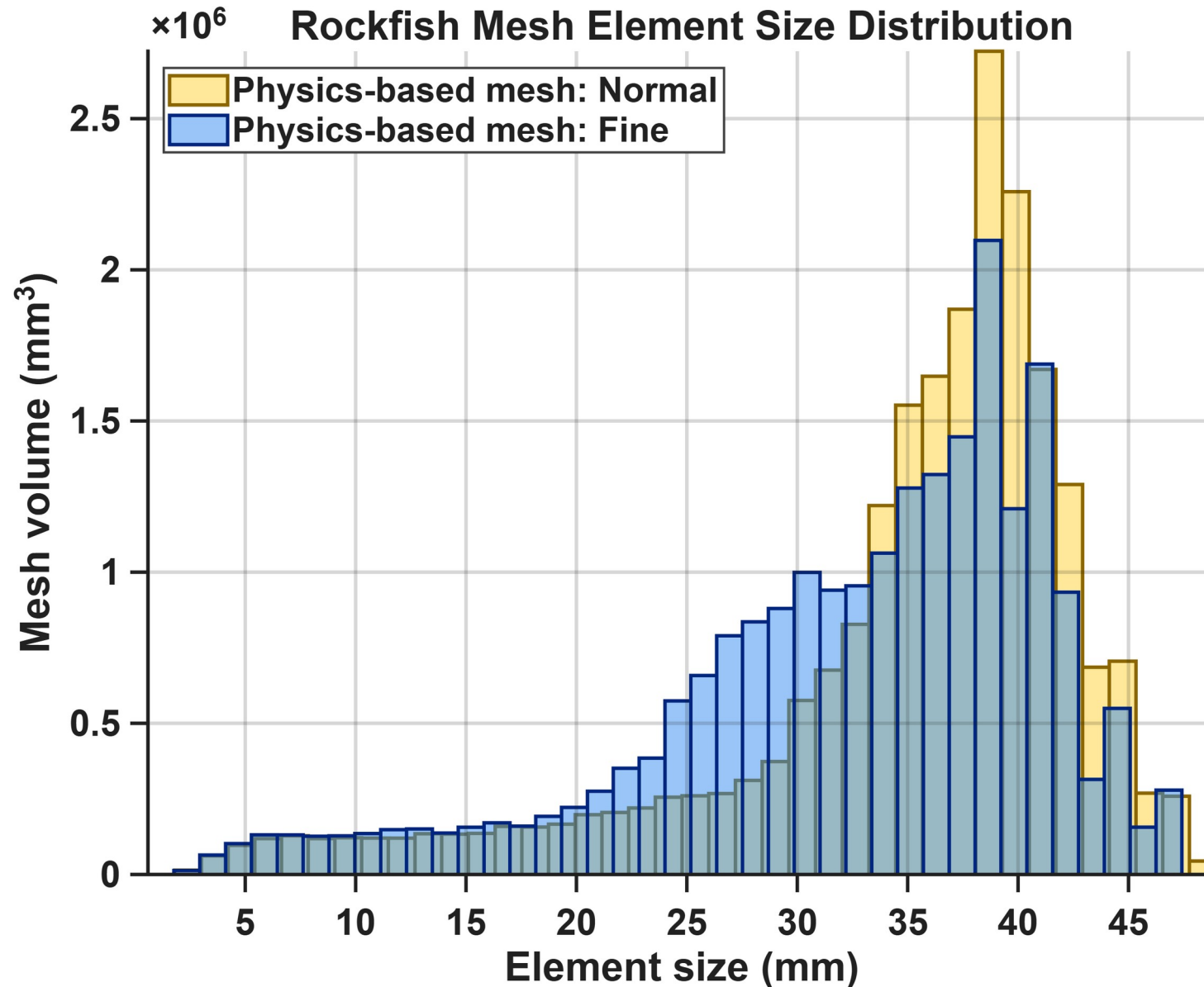


Mesh size comparisons

Soft sphere



Rockfish



NOTE

These distributions are for the mesh used for the highest frequency in the sweep.

I have not computed user-based mesh statistics for this one yet, but still interesting to see comparisons between fine and normal